

MECHANIC SETTLEMENT, NB, Canada

WMO: 717870

Lat: 45.6936N

Lon: 65.1650W

Elev: 1322

StdP: 14.01

Time Zone: -4.00 (NAA)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-3.2	-15.7	2.5	-5.9	-10.7	3.2	-0.5	27.0	24.3	25.2	21.7	13.1	290	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	78.8	66.6	76.1	65.1	73.6	63.4	69.2	75.0	67.7	72.7	66.2	70.6	9.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	105.9	70.7	66.0	100.7	69.6	64.4	95.1	68.3	34.1	75.2	32.9	72.8	31.6	70.5	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.4	20.9	18.8		DB	-13.9	84.4	4.3	2.5	-17.0	86.3	-19.6	87.8	-22.0	89.2	-25.2	91.0
				WB	-14.1	71.9	4.3	1.1	-17.2	72.6	-19.6	73.3	-22.0	73.9	-25.1	74.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	16.7	18.1	25.2	37.3	48.7	57.0	64.9	63.6	56.4	45.3	34.3	23.0
	DBStd	18.76	11.62	10.15	9.25	7.73	7.29	6.08	4.91	4.81	6.87	7.73	9.29	10.66
	HDD50	4727	1031	894	771	389	113	13	0	0	19	186	475	836
	HDD65	8874	1496	1314	1234	832	506	246	61	82	268	612	922	1301
	CDD50	1447	0	0	2	7	74	224	463	423	211	40	3	0
	CDD65	116	0	0	0	0	1	7	58	40	10	0	0	0
	CDH74	492	0	0	0	1	20	48	235	141	46	1	0	0
	CDH80	46	0	0	0	0	2	2	26	9	7	0	0	0
Wind	WSAvg	10.0	11.3	11.3	11.2	10.4	9.5	8.8	8.4	8.2	8.9	10.2	10.7	10.8
Precipitation	PrecAvg	51.8	5.0	3.4	4.8	4.0	4.4	3.7	3.6	3.0	4.6	5.2	4.9	5.0
	PrecMax	62.7	7.2	6.3	9.4	6.5	9.0	9.4	6.9	6.2	10.9	10.0	8.1	10.1
	PrecMin	38.6	2.6	1.1	0.8	2.0	1.8	1.5	1.6	0.6	1.7	1.1	2.6	1.8
	PrecStd	5.8	1.6	1.3	1.9	1.1	1.9	1.9	1.4	1.6	2.3	2.5	1.7	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	47.4	59.5	65.2	76.4	79.4	83.5	81.4	79.1	68.1	59.6	52.1
		MCWB	49.8	46.6	52.2	52.1	60.4	63.9	67.8	68.9	68.5	60.9	58.8	52.0
	2%	DB	43.6	40.8	45.0	57.7	70.2	74.1	79.1	77.8	73.7	63.3	56.0	46.5
		MCWB	43.3	39.5	41.1	49.7	56.7	61.7	67.1	66.6	65.2	58.6	54.6	45.6
	5%	DB	37.9	35.3	41.4	53.6	64.8	70.4	76.4	75.0	69.7	59.9	52.3	41.7
		MCWB	37.2	33.4	37.7	48.3	54.2	59.8	65.6	64.5	62.9	56.8	50.1	41.2
	10%	DB	32.8	32.2	37.6	50.0	60.8	67.5	74.0	72.2	66.4	56.8	48.4	37.5
		MCWB	32.1	31.1	34.1	44.7	51.8	57.6	64.3	63.2	61.8	53.5	46.2	36.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	46.9	52.2	57.3	63.8	67.6	71.2	71.0	69.5	63.1	59.4	52.1
		MCDB	48.8	47.0	59.3	58.0	72.6	74.3	77.3	77.4	77.5	64.5	59.5	52.1
	2%	WB	42.8	39.9	43.1	53.1	60.4	64.8	69.3	69.0	66.8	60.4	55.6	46.1
		MCDB	43.1	40.4	44.3	55.5	65.6	69.1	75.5	74.0	70.9	61.5	55.8	46.4
	5%	WB	36.6	34.2	38.6	49.7	57.4	62.7	68.1	67.2	65.1	57.4	51.0	41.3
		MCDB	37.0	35.1	39.9	52.0	61.9	67.3	73.2	72.0	67.8	59.2	51.5	41.5
	10%	WB	31.9	31.2	34.3	45.3	54.3	60.4	66.5	65.5	63.2	54.1	46.5	37.0
		MCDB	32.6	31.8	36.9	48.8	59.3	64.9	71.3	70.0	65.7	56.2	47.9	37.6
Mean Daily Temperature Range	5% DB	MDBR	15.0	14.6	14.6	14.6	16.9	15.0	14.6	14.4	14.6	12.9	12.2	13.2
		MCDBR	20.6	17.1	18.1	20.0	22.0	19.4	17.8	17.2	17.2	15.2	16.3	17.8
		MCWBR	20.3	15.3	14.6	14.3	12.8	10.7	9.3	9.6	10.4	11.7	15.4	18.1
	5% WB	MCDBR	19.7	17.1	17.4	18.1	19.0	16.6	15.4	15.0	14.7	14.6	16.1	17.6
		MCWBR	20.4	16.6	16.7	15.5	13.5	10.9	9.3	9.6	10.9	13.1	16.6	18.2
Clear-Sky Solar Irradiance	taub	0.274	0.274	0.305	0.331	0.353	0.378	0.405	0.375	0.342	0.313	0.295	0.275	
	taud	2.437	2.437	2.443	2.431	2.408	2.356	2.312	2.398	2.475	2.558	2.537	2.494	
	Ebn at Noon	268	292	295	295	290	282	272	277	278	274	259	256	
	Edh at Noon	22	26	30	34	36	38	39	35	30	23	20	19	
All-Sky Solar Radiation	RadAvg	459	721	1090	1361	1582	1711	1756	1564	1247	794	443	344	
	RadStd	41	56	69	128	164	142	117	107	96	82	49	27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	+86	N/A

Nomenclature: See separate page